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Proning frame for a patient bed

Abstract

A head rest accessory for coupling to a frame member at a head end of a pivotable torso support section of a hospital bed is provided. A frame is configured to couple to the frame member. A head rest mount is coupled to the frame and is repositionable along the frame. A head rest is coupled to the head rest mount and includes a face rest shell. A first angle sensor is positioned along a side of the face rest shell. A second angle sensor is positioned on a head end of the face rest shell.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 63/105,094, filed Oct. 23, 2020, and U.S. Provisional Patent Application Ser. No. 63/155,827, filed Mar. 3, 2021, both of which are expressly incorporated by reference herein.

BACKGROUND

(1) The present disclosure relates to hospital beds that enable proning of a patient and, more particularly, to a head rest attachment for a hospital bed that enables the patient to lie in the prone position.

(2) Proning is the process of turning a patient with precise, safe motions from their back onto their abdomen (stomach) so that the patient is lying face down. Proning is used to improve the respiratory function in high risk patients. During proning, the patient is placed in prone position for extended amounts of time. The patient's head must be supported during proning, and the patient cannot be placed face down on a mattress. Accordingly, providing comfort for the patient is desirable. Additionally, during proning, caregivers need to have access to the patient's face for intubation purposes.

(3) Generally, C-Prone masks do not include tilt indicators. Accordingly it is difficult for a caregiver to know an exact position of the mask when adjusting the mask after positioning a patient's head on the mask. If the mask has a lateral tilt, an undesirable pressure interface distribution may occur on the patient's face, thereby producing skin injuries. Without a longitudinal tilt indicator, the caregiver cannot improve the patient's head positioning to accommodate patient morphology after observation of the posture of the patient's head and neck. Moreover, without tilt indicators, it is not possible to pre-set a mask position before fixing the mask on the bed. Accordingly, all position setting must be done after the patient's head is positioned on the mask.

SUMMARY

(4) The present disclosure includes one or more of the features recited in the appended claims and/or the following features which, alone or in any combination, may comprise patentable subject matter.

(5) According to a first aspect of the disclosed embodiments, a head rest accessory for coupling to a frame member at a head end of a pivotable torso support section of a hospital bed includes a frame having a pair of rails that are spaced apart and substantially parallel and that extend from a distal end to a proximal end. A crossbeam is coupled to the distal ends of the pair of rails. A U-shaped connector is coupled to the proximal ends of a respective rail of the pair of rails. The U-shaped connector opens downwardly to permit the U-shaped connector to be insertable downwardly over the frame member to couple the frame to the pivotable torso support section so that the frame is cantilevered from the frame member. A head rest mount is coupled to the frame and extends between the pair of rails. The head rest mount has at least one releasable clamp that grips a corresponding one of the pair of rails to lock the head rest mount in place on the frame when the releasable clamp is in a locked condition. The head rest mount is repositionable along the pair of rails when the releasable clamp is in an unlocked condition. A head rest is coupled to the head rest mount. The head rest includes a face rest shell having an upwardly facing concave surface and having a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(6) In some embodiments of the first aspect, the crossbeam may extend orthogonal to each of the pair of rails. The U-shaped connector may extend perpendicular to the frame. The U-shaped connector may include a pair of flanges and a slot extending between the pair of flanges. Each of the pair of flanges may include a planar surface. The slot may be defined by the planar surfaces of the pair of flanges. The crossbeam may include a rod. Each of the pair of rails may have a rectangular cross-section taken in a vertical plane. Each of the pair of rails may include a loop configured to position over a base frame of the bed when the bed is in a Trendelenburg position. The U-shaped connector may include a pair of sockets. Each loop may include a post configured to be inserted into a respective socket. The posts may be removably inserted into each of the pair of sockets. At least a portion of the U-shaped connector may include at least one of a plastic and composite.

(7) Optionally, in the first aspect, the head rest mount may be repositionable along the pair of rails along a longitudinal axis of the hospital bed. The head rest mount may include a base that is repositionable along the pair of rails. The base may include a track and a body repositionable along the track in a direction perpendicular to the longitudinal axis of the hospital bed. A post may extend vertically from the body. A mount rail may extend from the post and repositionable vertically along the post. The head rest may be coupled to a cantilevered end of the rail.

(8) It may be desired, in the first aspect, that a face cushion is positioned on the concave surface of the face rest shell. The face cushion may include a surface that is sized and shaped to the patient's face. The face cushion may include a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(9) According to a second aspect of the disclosed embodiments, a head rest accessory for coupling

to a frame member at a head end of a pivotable torso support section of a hospital bed includes a frame having a pair of rails that are spaced apart and substantially parallel and that extend from a distal end to a proximal end. A crossbeam is coupled to the distal ends of the pair of rails. A U-shaped connector is coupled to and extends orthogonally from the proximal ends of a respective rail of the pair of rails. The U-shaped connector opens downwardly to permit the U-shaped connector to be insertable downwardly over the frame member to couple the frame to the pivotable torso support section so that the frame is cantilevered from the frame member. A head rest mount is coupled to the frame and extends between the pair of rails. The head rest mount has at least one releasable clamp that grips a corresponding one of the pair of rails to lock the head rest mount in place on the frame when the releasable clamp is in a locked condition. The head rest mount is repositionable along the pair of rails along a longitudinal axis of the hospital bed when the releasable clamp is in an unlocked condition. A head rest is coupled to the head rest mount. The head rest includes a face rest shell having an upwardly facing concave surface and having a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(10) In some embodiments of the second aspect, the crossbeam may extend orthogonal to each of the pair of rails. The U-shaped connector may extend perpendicular to the frame. The U-shaped connector may include a pair of flanges and a slot extending between the pair of flanges. Each of the pair of flanges may include a planar surface. The slot may be defined by the planar surfaces of the pair of flanges. The crossbeam may include a rod. Each of the pair of rails may have a rectangular cross-section taken in a vertical plane. Each of the pair of rails may include a loop configured to position over a base frame of the bed when the bed is in a Trendelenburg position. The U-shaped connector may include a pair of sockets. Each loop may include a post configured to be inserted into a respective socket. The posts may be removably inserted into each of the pair of sockets. At least a portion of the U-shaped connector may include at least one of a plastic and composite.

(11) Optionally, in the second aspect, the head rest mount may include a base that is repositionable along the pair of rails. The base may include a track and a body repositionable along the track in a direction perpendicular to the longitudinal axis of the hospital bed. A post may extend vertically from the body. A mount rail may extend from the post and repositionable vertically along the post. The head rest may be coupled to a cantilevered end of the rail.

(12) It may be desired, in the second aspect, that a face cushion is positioned on the concave surface of the face rest shell. The face cushion may include a surface that is sized and shaped to the patient's face. The face cushion may include a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(13) According to a third aspect of the disclosed embodiments, a head rest accessory for coupling to a frame member at a head end of a pivotable torso support section of a hospital bed includes a C-shaped frame having a pair of rails that are spaced apart and substantially parallel and that extend from a distal end to a proximal end. A crossbeam is coupled to the distal ends of the pair of rails. A U-shaped connector is coupled to the proximal ends of a respective rail of the pair of rails. The U-shaped connector includes a pair of flanges having planar surfaces and a slot extending between the pair of flanges. The slot is defined by the planar surfaces of the pair of flanges. The U-shaped connector opens downwardly to permit the U-shaped connector to be insertable downwardly over the frame member to couple the frame to the pivotable torso support section so that the frame is cantilevered from the frame member. A head rest mount is coupled to the frame and extends between the pair of rails. The head rest mount has at least one releasable clamp that grips a corresponding one of the pair of rails to lock the head rest mount in place on the frame when the releasable clamp is in a locked condition. The head rest mount is repositionable along the pair of rails when the releasable clamp is in an unlocked condition. A head rest is coupled to the head rest

mount. The head rest includes a face rest shell having an upwardly facing concave surface and having a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(14) In some embodiments of the third aspect, the crossbeam may extend orthogonal to each of the pair of rails. The U-shaped connector may extend perpendicular to the frame. The U-shaped connector may include a pair of flanges and a slot extending between the pair of flanges. Each of the pair of flanges may include a planar surface. The slot may be defined by the planar surfaces of the pair of flanges. The crossbeam may include a rod. Each of the pair of rails may have a rectangular cross-section taken in a vertical plane. Each of the pair of rails may include a loop configured to position over a base frame of the bed when the bed is in a Trendelenburg position. The U-shaped connector may include a pair of sockets. Each loop may include a post configured to be inserted into a respective socket. The posts may be removably inserted into each of the pair of sockets. At least a portion of the U-shaped connector may include at least one of a plastic and composite.

(15) Optionally, in the third aspect, the head rest mount may be repositionable along the pair of rails along a longitudinal axis of the hospital bed. The head rest mount may include a base that is repositionable along the pair of rails. The base may include a track and a body repositionable along the track in a direction perpendicular to the longitudinal axis of the hospital bed. A post may extend vertically from the body. A mount rail may extend from the post and repositionable vertically along the post. The head rest may be coupled to a cantilevered end of the rail.

(16) It may be desired, in the third aspect, that a face cushion is positioned on the concave surface of the face rest shell. The face cushion may include a surface that is sized and shaped to the patient's face. The face cushion may include a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(17) According to a fourth aspect of the disclosed embodiment, a head rest accessory for coupling to a frame member at a head end of a pivotable torso support section of a hospital bed includes a frame having a pair of rails that are spaced apart and substantially parallel and that extend from a distal end to a proximal end. A crossbeam is coupled to the distal ends of the pair of rails. A U-shaped connector is coupled to the proximal ends of a respective rail of the pair of rails. The U-shaped connector opens downwardly to permit the U-shaped connector to be insertable downwardly over the frame member to couple the frame to the pivotable torso support section so that the frame is cantilevered from the frame member. A head rest mount is coupled to the frame and extends between the pair of rails. The head rest mount has at least one releasable clamp that grips a corresponding one of the pair of rails to lock the head rest mount in place on the frame when the releasable clamp is in a locked condition. The head rest mount is repositionable along the pair of rails when the releasable clamp is in an unlocked condition. A head rest is coupled to the head rest mount. The head rest includes a face rest shell having an upwardly facing concave surface and having a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory. The head rest includes a face cushion positioned on the face rest shell and moveable relative to the frame in three dimensions.

(18) In some embodiments of the fourth aspect, the crossbeam may extend orthogonal to each of the pair of rails. The U-shaped connector may extend perpendicular to the frame. The U-shaped connector may include a pair of flanges and a slot extending between the pair of flanges. Each of the pair of flanges may include a planar surface. The slot may be defined by the planar surfaces of the pair of flanges. The crossbeam may include a rod. Each of the pair of rails may have a rectangular cross-section taken in a vertical plane. Each of the pair of rails may include a loop configured to position over a base frame of the bed when the bed is in a Trendelenburg position. The U-shaped connector may include a pair of sockets. Each loop may include a post configured to

be inserted into a respective socket. The posts may be removably inserted into each of the pair of sockets. At least a portion of the U-shaped connector may include at least one of a plastic and composite.

(19) Optionally, in the fourth aspect, the head rest mount may be repositionable along the pair of rails in a first dimension along a longitudinal axis of the hospital bed. The head rest mount may include a base that is repositionable in the first dimension along the pair of rails. The base may include a track and a body repositionable along the track in a second dimension perpendicular to the longitudinal axis of the hospital bed. A post may extend vertically from the body. A mount rail may extend from the post and repositionable in a third dimension along the post. The head rest may be coupled to a cantilevered end of the rail.

(20) It may be desired, in the fourth aspect, that the face cushion is positioned on the concave surface of the face rest shell. The face cushion may include a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(21) According to any of the aspects set forth above, the head rest may be rotatable about the mount rail in a first rotational direction to roll the head rest. The head rest may be rotatable about the mount rail in a second rotational direction to pitch the head rest. The head rest may be rotatable about the mount rail in a third rotational direction to yaw the head rest.

(22) According to any of the aspects set forth above, the face rest shell may include a downwardly facing convex surface opposite the upwardly facing concave surface. An outer perimeter may have a partially circular top outer edge and a bottom outer edge. The outer perimeter may include a pair of planar side outer edges extending between the top outer edge and the bottom outer edge. A bottom opening may be formed in the bottom outer edge. The generally T-shaped cutout may extend from the bottom opening. The generally T-shaped cutout may be defined by an inner edge. The generally T-shaped cutout may include a mouth cutout extending from the opening and an eye cutout extending from the mouth cutout. The eye cutout may be defined by a planar top inner edge and a pair of curved inner edges extending from the planar top inner edge. The concave surface and the convex surface may extend between the inner edge of the cutout and the outer perimeter.

(23) In some embodiments of any of the above aspects, the bottom outer edge may include a pair of planar bottom outer edges. The opening may be formed between the pair of bottom outer edges. Each of a pair of planar angled outer edges may extend between one of the pair of planar side outer edges and one of the planar bottom outer edges. The planar top inner edge of the eye cutout may extend substantially perpendicular to the pair of planar side outer edges of the outer perimeter. The planar top inner edge of the eye cutout may extend substantially parallel to each of the pair of planar bottom outer edges. The pair of curved inner edges of the eye cutout may be at least partially circular. The pair of curved inner edges of the eye cutout may be at least partially arcuate.

(24) Optionally, in any of the above aspects, the face rest shell may include a pair of cheek rests extending between the eye cutout and the mouth cutout. A cheek rest opening may extend between an inwardly most point of each of the pair of cheek rests. The bottom opening may have a first width and the cheek rest opening has a second width that is less than the first width. The bottom opening may have a first width and the cheek rest opening has a second width that is substantially equal to the first width. Each cheek rest may include a cheek rest relief surface formed in the upwardly facing concave surface. The cheek rest relief surface may be a concave surface having a radius of curvature that differs from a radius of curvature of the upwardly facing concave surface. A perimeter of each cheek rest relief surface may be defined by a portion of the respective curved inner edge of the eye cutout and a curved bottom relief edge.

(25) It may be desired in any of the above aspects that the mouth cutout includes a pair of side mouth inner edges extending from the bottom opening. The pair of side mouth inner edges may be at least partially arcuate. The pair of side mouth inner edges may each include at least one planar side mouth inner edge. Each of the pair of side inner mouth edges may extend from the bottom

outer edge of the outer perimeter. The outer perimeter may include a curved outer edge coupling the top outer edge and the pair of planar side outer edges.

(26) According to a fifth aspect of the disclosed embodiments, a face rest shell for a head rest accessory includes an upwardly facing concave surface. A downwardly facing convex surface is opposite the upwardly facing concave surface. An outer perimeter has a partially circular top outer edge and a bottom outer edge. The outer perimeter includes a pair of planar side outer edges extending between the top outer edge and the bottom outer edge. A bottom opening is formed in the bottom outer edge. A cutout extends from the bottom opening. The cutout is configured to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory. The cutout is defined by an inner edge. The cutout includes a mouth cutout extending from the opening and an eye cutout extending from the mouth cutout. The eye cutout is defined by a planar top inner edge and a pair of curved inner edges extending from the planar top inner edge. The concave surface and the convex surface extend between the outer edge of the cutout and the outer perimeter.

(27) In some embodiments of the fifth aspect, the bottom outer edge may include a pair of planar bottom outer edges. The opening may be formed between the pair of bottom outer edges. Each of a pair of planar angled outer edges may extend between one of the pair of planar side outer edges and one of the planar bottom outer edges. The planar top inner edge of the eye cutout may extend substantially perpendicular to the pair of planar side outer edges of the outer perimeter. The planar top inner edge of the eye cutout may extend substantially parallel to each of the pair of planar bottom outer edges. The pair of curved inner edges of the eye cutout may be at least partially circular. The pair of curved inner edges of the eye cutout may be at least partially arcuate.

(28) Optionally, in the fifth aspect, the face rest shell may include a pair of cheek rests extending between the eye cutout and the mouth cutout. A cheek rest opening may extend between an inwardly most point of each of the pair of cheek rests. The bottom opening may have a first width and the cheek rest opening has a second width that is less than the first width. The bottom opening may have a first width and the cheek rest opening has a second width that is substantially equal to the first width. Each cheek rest may include a cheek rest relief surface formed in the upwardly facing concave surface. The cheek rest relief surface may be a concave surface having a radius of curvature that differs from a radius of curvature of the upwardly facing concave surface. A perimeter of each cheek rest relief surface may be defined by a portion of the respective curved inner edge of the eye cutout and a curved bottom relief edge.

(29) It may be desired in the fifth aspect that the mouth cutout includes a pair of side mouth inner edges extending from the bottom opening. The pair of side mouth inner edges may be at least partially arcuate. The pair of side mouth inner edges may each include at least one planar side mouth inner edge. Each of the pair of side inner mouth edges may extend from the bottom outer edge of the outer perimeter. The outer perimeter may include a curved outer edge coupling the top outer edge and the pair of planar side outer edges. The cutout may be generally T-shaped.

(30) In some embodiments of any of the above aspects, the face rest shell may include at least one angle sensor attached to the face rest shell to indicate a tilt angle of the face rest shell. The at least one angle sensor may be positioned along a side of the face rest shell and aligned with an eye cutout of the generally T-shaped cutout. The at least one angle sensor may measure a longitudinal tilt angle of the face rest shell. The at least one angle sensor may be positioned on a head end of the face rest shell on a centerline of the face rest shell extending between the head end and a chin end of the face rest shell. The at least one angle sensor may measure a lateral tilt angle of the face rest shell. The at least one angle sensor may include a first angle sensor positioned along a side of the face rest shell and aligned with an eye cutout of the generally T-shaped cutout to measure a longitudinal tilt angle of the face rest shell. A second angle sensor may be positioned on a head end of the face rest shell on a centerline of the face rest shell extending between the head end and a chin end of the face rest shell to measure a lateral tilt angle of the face rest shell.

(31) Optionally, the at least one angle sensor may include a spirit level vial having a bubble that indicates the angle of the face rest shell. The at least one angle sensor may include a liquid level that indicates the angle of the face rest shell. The liquid level may include an artificial horizon inclinometer. The artificial horizon inclinometer may indicate one of a lateral tilt angle or a longitudinal tilt angle of the face rest shell. The artificial horizon inclinometer may indicate both of a lateral tilt angle and a longitudinal tilt angle of the face rest shell. The at least one angle sensor may include a plurality of markings indicative of an angle of the face rest shell.

(32) It may be contemplated that the at least one angle sensor may facilitate setting a correct position of the face rest shell according a patient morphology. The at least one angle sensor may facilitate setting a correct position of the face rest shell before the patient's face is positioned in the face rest shell.

(33) According to a sixth aspect of the disclosed embodiments, a head rest accessory is provided for coupling to a frame member at a head end of a pivotable torso support section of a hospital bed. The head rest accessory includes a frame configured to couple to the frame member so that the frame is cantilevered from the frame member. A head rest mount is coupled to the frame and is repositionable along the frame. A head rest is coupled to the head rest mount. The head rest includes a face rest shell having an upwardly facing concave surface and having a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory. A first angle sensor is positioned along a side of the face rest shell and aligned with an eye cutout of the generally T-shaped cutout to measure a first tilt angle of the face rest shell. A second angle sensor is positioned on a head end of the face rest shell on a centerline of the face rest shell extending between the head end and a chin end of the face rest shell to measure a second tilt angle of the face rest shell.

(34) In some embodiments of the sixth aspect, the first tilt angle may include a longitudinal tilt angle. The second tilt angle may include a lateral tilt angle. At least one of the first angle sensor and the second angle sensor may include a spirit level vial having a bubble that indicates the angle of the face rest shell. At least one of the first angle sensor and the second angle sensor may include a liquid level that indicates the angle of the face rest shell. The liquid level may include an artificial horizon inclinometer. The artificial horizon inclinometer may indicate one of a lateral tilt angle or a longitudinal tilt angle of the face rest shell. The artificial horizon inclinometer may indicate both of a lateral tilt angle and a longitudinal tilt angle of the face rest shell. At least one of the first angle sensor and the second angle sensor may include a plurality of markings indicative of an angle of the face rest shell.

(35) Optionally, in the sixth aspect, at least one of the first angle sensor and the second angle sensor may facilitate setting a correct position of the face rest shell according a patient morphology. At least one of the first angle sensor and the second angle sensor may facilitate setting a correct position of the face rest shell before the patient's face is positioned in the face rest shell.

(36) According to a seventh aspect of the disclosed embodiments, a head rest accessory for coupling to a frame member at a head end of a pivotable torso support section of a hospital bed is provided. The head rest accessory includes a frame having a pair of rails that are spaced apart and substantially parallel and that extend from a distal end to a proximal end. A crossbeam is coupled to the distal ends of the pair of rails. A U-shaped connector is coupled to the proximal ends of a respective rail of the pair of rails. The U-shaped connector opens downwardly to permit the U-shaped connector to be insertable downwardly over the frame member to couple the frame to the pivotable torso support section so that the frame is cantilevered from the frame member. A head rest mount is coupled to the frame and extends between the pair of rails. The head rest mount has at least one releasable clamp that grips a corresponding one of the pair of rails to lock the head rest mount in place on the frame when the releasable clamp is in a locked condition. The head rest mount is repositionable along the pair of rails when the releasable clamp is in an unlocked condition. A head rest is coupled to the head rest mount. The head rest includes a face rest shell

having an upwardly facing concave surface and having a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(37) In some embodiments of the seventh aspect, the crossbeam may extend orthogonal to each of the pair of rails. The U-shaped connector may extend perpendicular to the frame and may include a pair of flanges and a slot extending between the pair of flanges. Each of the pair of flanges may include a planar surface. The slot may be defined by the planar surfaces of the pair of flanges. Each of the pair of rails may have a rectangular cross-section taken in a vertical plane. Each of the pair of rails may include a loop configured to position over a base frame of the bed when the bed is in a Trendelenburg position. The U-shaped connector may include a pair of sockets. Each loop may include a post configured to be inserted into a respective socket. The posts may be removably inserted into each of the pair of sockets. At least a portion of the U-shaped connector may include at least one of a plastic and composite.

(38) Optionally, in the seventh aspect, the head rest mount may be repositionable along the pair of rails along a longitudinal axis of the hospital bed. The head rest mount may include a base that is repositionable along the pair of rails. The base may include a track and a body repositionable along the track in a direction perpendicular to the longitudinal axis of the hospital bed. A post may extend vertically from the body. A mount rail may extend from the post and repositionable vertically along the post. The head rest may be coupled to a cantilevered end of the rail. The head rest may be rotatable about the mount rail in a first rotational direction to roll the head rest. The head rest may be rotatable about the mount rail in a second rotational direction to pitch the head rest. The head rest may be rotatable about the mount rail in a third rotational direction to yaw the head rest.

(39) It may be desired, in the seventh aspect, that the face rest shell includes a downwardly facing convex surface opposite the upwardly facing concave surface. An outer perimeter may have a partially circular top outer edge and a bottom outer edge. The outer perimeter may include a pair of planar side outer edges extending between the top outer edge and the bottom outer edge. A bottom opening may be formed in the bottom outer edge. The generally T-shaped cutout may extend from the bottom opening. The generally T-shaped cutout may be defined by an inner edge. The generally T-shaped cutout may include a mouth cutout extending from the opening and an eye cutout extending from the mouth cutout. The eye cutout may be defined by a planar top inner edge and a pair of curved inner edges extending from the planar top inner edge. The concave surface and the convex surface may extend between the inner edge of the cutout and the outer perimeter. The face rest shell may include relief cutouts around the eye cutout to facilitate distributing interface pressure from protruding facial features of the patient. The face rest shell may include a top wall extending from the top outer edge. A pair of side walls may be provided. Each of the pair of side walls may extend from one of the pair of planar side outer edges. A face cushion may be positioned on the concave surface of the face rest shell and may be retained in the concave surface of the face rest shell by the top wall and the pair of side walls. The face cushion may include a surface that is sized and shaped to the patient's face. The face cushion may include a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

(40) In some embodiments of the seventh aspect, the face rest shell may include at least one angle sensor attached to the face rest shell to indicate a tilt angle of the face rest shell. The at least one angle sensor may be positioned along a side of the face rest shell and aligned with an eye cutout of the generally T-shaped cutout. The at least one angle sensor may measure a longitudinal tilt angle of the face rest shell. The at least one angle sensor may be positioned on a head end of the face rest shell on a centerline of the face rest shell extending between the head end and a chin end of the face rest shell. The at least one angle sensor may measure a lateral tilt angle of the face rest shell.

(41) Additional features, which alone or in combination with any other feature(s), such as those listed above and those listed in the claims, may comprise patentable subject matter and will become

apparent to those skilled in the art upon consideration of the following detailed description of various embodiments exemplifying the best mode of carrying out the embodiments as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The detailed description particularly refers to the accompanying figures in which:
- (2) FIG. 1 is a perspective view of a patient support apparatus illustrated as a hospital bed;
- (3) FIG. 2 is a head end view of the patient support apparatus shown in FIG. 1;
- (4) FIG. 3 is a perspective view of a head rest accessory formed in accordance with an embodiment;
- (5) FIG. 4 is a perspective view of the head rest accessory shown in FIG. 3 coupled to the patient support apparatus shown in FIG. 1;
- (6) FIG. 5 is a perspective view of the head rest accessory shown in FIG. 3 having a head rest mount and face rest shell coupled thereto;
- (7) FIG. 6 is a perspective view of a face cushion positioned on the face rest shell shown in FIG. 5;
- (8) FIG. 7 is a front view of a face cushion formed in accordance with an embodiment;
- (9) FIG. 8 is a front view of a patient's face positioned in the face cushion shown in FIG. 7, wherein the face cushion is positioned in the face rest shell shown in FIG. 5;
- (10) FIG. 9 is a perspective view of the head rest accessory shown in FIG. 3 coupled to the patient support apparatus shown in FIG. 1 with the patient support apparatus in a Trendelenburg position.
- (11) FIG. 10 is a front view of a face rest shell formed in accordance with an embodiment and showing a downwardly facing convex surface;
- (12) FIG. 11 is a rear view of the face rest shell shown in FIG. 10 and showing an upwardly facing concave surface;
- (13) FIG. 12 is a front view of a face rest shell formed in accordance with another embodiment and showing a downwardly facing convex surface;
- (14) FIG. 13 is a rear view of the face rest shell shown in FIG. 12 and showing an upwardly facing concave surface;
- (15) FIG. 14 is a front view of a face rest shell formed in accordance with yet another embodiment and showing a downwardly facing convex surface;
- (16) FIG. 15 is a rear view of the face rest shell shown in FIG. 14 and showing an upwardly facing concave surface;
- (17) FIG. 16 is a front view of a face rest shell formed in accordance with a further embodiment and showing a downwardly facing convex surface;
- (18) FIG. 17 is a rear view of the face rest shell shown in FIG. 16 and showing an upwardly facing concave surface;
- (19) FIG. 18 is a head end view of a patient support apparatus illustrating a lateral angle of a face rest shell;
- (20) FIG. 19 is a side view of a patient support apparatus illustrating a longitudinal angle of a face rest shell;
- (21) FIG. 20 is a perspective view of an embodiment of a face rest shell having a lateral angle sensor and a longitudinal angle sensor;
- (22) FIG. 21 is a front view of the lateral angle sensor of the face rest shell of FIG. 20;
- (23) FIG. 22 is a side view of the longitudinal angle sensor of the face rest shell of FIG. 20;
- (24) FIG. 23 is a perspective view of another embodiment of a face rest shell having a lateral angle sensor and a longitudinal angle sensor;
- (25) FIG. 24 is a front view of the lateral angle sensor of the face rest shell of FIG. 23;

- (26) FIG. 25 is a side view of the longitudinal angle sensor of the face rest shell of FIG. 23;
- (27) FIG. 26 is a perspective view of yet another embodiment of a face rest shell having a lateral angle sensor and a longitudinal angle sensor;
- (28) FIG. 27 is a front view of the lateral angle sensor of the face rest shell of FIG. 26;
- (29) FIG. 28 is a side view of the longitudinal angle sensor of the face rest shell of FIG. 26;
- (30) FIG. 29 is a front view of a face rest shell formed in accordance with a further embodiment and showing a downwardly facing convex surface;
- (31) FIG. 30 is a rear view of the face rest shell shown in FIG. 29 and showing an upwardly facing concave surface; and
- (32) FIG. 31 is a front view of a face cushion formed in accordance with an embodiment.

DETAILED DESCRIPTION

(33) While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific exemplary embodiments thereof have been shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

(34) Referring to FIG. 1, a patient support apparatus 12 includes a base 34 and a deck 36 that support a patient support surface 14 above the floor. The base 34 is configured to raise and lower the deck 36 relative to the floor to raise and lower the patient support surface 14 relative to the floor. The deck 36 is articulatable and may be reconfigured to support a patient on the patient support surface 14 in a variety of positions, for example in a lie-flat position or a sit-up position (shown in FIG. 1). The patient support apparatus 12 also includes siderails 38 and headrails 40 coupled to the deck 36 to block a patient from accidentally rolling off of the patient support system 12.

(35) The base 34 illustratively includes a lower frame 42, an upper frame 44, and a lift system 22. The lift system 22 is illustratively coupled between the lower frame 42 and the upper frame 44 to raise and lower the upper frame 44 relative to the lower frame 42. The lift system 22 illustratively includes lift arms 61 that pivot relative to the lower frame 42 and the upper frame 44 to raise and lower the upper frame 44 relative to the lower frame 42.

(36) The deck 36 illustratively includes a torso support section 102, a seat-deck section 48, and a foot-deck section 52. The torso support section 102 is mounted to the upper frame 44 to pivot about an axis relative to the seat-deck section 48. The seat-deck section 48 is coupled to the upper frame 44 to move with the upper frame 44. The foot-deck section 52 is coupled to the seat-deck section 48 to pivot relative to the seat-deck section 48. The foot-deck section 52 is also extendable and retractable to lengthen or shorten the deck 36 as desired by a caregiver or to accommodate repositioning of the deck 36.

(37) Referring to FIG. 2, the apparatus 12 includes the upper frame 44 at a head end 100 of a pivotable torso support section 102. The upper frame 44 includes a crossbar 104 that is generally rectangular in shape. The crossbar 104 extends along the head end 100 of the support section 102. A panel 110 is positioned in the upper frame 44 and includes a head end 112 that extends along the crossbar 104. The panel 110 is spaced from the crossbar 104 to create a slot 114 between the head end 112 of the panel 110 and the crossbar 104, in some embodiments. In other embodiments, the panel 110 includes cutout (not shown) to create the slot 114 between the head end 112 of the panel 110 and the crossbar 104. A lower frame member 120 is positioned below the frame 44. The lower frame member 120 is positioned outward in a head end direction from the crossbar 104. The lower frame member 120 includes sockets 122 for attaching accessories to the apparatus 12.

(38) Referring now to FIG. 3, a head rest accessory 130 includes a frame 132 having a pair of rails 134 that extend from a proximal end 136 to a distal end 138. The frame 132 is a substantially C-shaped frame 132 and the pair of rails 134 extend substantially parallel to one another along a

respective longitudinal axis **140**. A crossbeam **150** extends between and is coupled to each of the pair of rails **134** at the distal end **138**. In the illustrated embodiment, the crossbeam **150** is configured as a rod. The crossbeam **150** extends orthogonal to the longitudinal axis **140** of each of the pair of rails **134**. That is, the crossbeam **150** extends perpendicular to the longitudinal axis **140** of each of the pair of rails **134**. Each rail **134** includes a rectangular member **135** that is formed from a plastic or composite. In some embodiments, the members **135** may be formed from metal or any other suitable material. In some embodiments, the members **135** have a relatively rectangular cross-section in a vertical plane **156**. In other embodiments, the members **135** may have any suitable cross-section. In some embodiments, the cross-sectional shape of the members **135** is in accordance with the longitudinal frame members of the Allen Advance® Table. In some embodiments, the members **135** have the dimensions of approximately 1.5 inches (about 3.81 centimeters) high and approximately 1.25 inches (about 3.175 centimeters) wide. In some embodiments, the spacing between the members **135** is approximately 14.5 inches (about 36.83 centimeters).

(39) A U-shaped connector **160** extends between the proximal end **136** of each rail **134**. The connector **160** extends orthogonal to each of the rails **134**. That is, the connector **160** extends substantially perpendicular to the rails **134**. The connector **160** includes a pair of spaced apart flanges **162** that define a slot **164** between the flanges **162**. Each flange **162** includes a planar surface **166** facing inwardly to define a respective surface of the slot **164**. The planar surface **166** extends from an end wall **168** of the slot **164** to an opening **170** of the slot **164**. In some embodiments, at least a portion of the connector **160** is formed from a plastic or composite. For example, an inner portion **168** of each flange **162** that forms the planar surfaces **166** and the end wall **168** may be formed from a plastic or composite. In some embodiments, the entire connector **160** is formed from a plastic or composite. The slot **164** opens downwardly so that the opening **170** is positioned at a bottom end **172** of the connector **162** and the end wall **168** is positioned above the opening **170**. The slot **164** extends substantially orthogonally from the longitudinal axis **140** of each rail **134**. More specifically, the slot **164** extends perpendicular to the longitudinal axis **140** of each rail **134**. The connector **160** is configured to be insertable downwardly over the crossbeam **104** to couple the head rest accessory **130** to the pivotable torso support section **102**.

(40) Each rail **134** includes a loop configuration **190** at the proximal end **136**. The loop configuration **190** includes a pair of outwardly and upwardly extending members **191**. Each of the members **191** is coupled together by an upper rail **192**. The upper rail **192** extends substantially parallel to the respective rail **134**. The upper rail **192** is positioned above the respective rail **134** so that a space **193** is defined in the loop configuration **190**. A post **194** extends downwardly from each loop configuration **190**. The connector **160** includes a pair of sockets **195** that are each configured to receive one of the posts **194** so that the rails **134** are coupled to the connector **160**. In some embodiments, the rails **134** are removably coupled to the connector **160**. That is, the posts **194** are removable from the sockets **195** to disconnect the rails **134** from the connector **160**. In this way, the rails **134** may be removed to allow room for various accessories without removing the connector **160** from the apparatus **12**.

(41) Referring now to FIG. 4, the head rest accessory **130** is illustrated coupled to the crossbar **104** of the apparatus **12**. The connector **160** of the head rest accessory **130** is positioned downwardly onto the crossbar **104** of the upper frame **44** so that the crossbar **104** is received within the slot **164** of the connector **160**. The head rest accessory **130** is coupled to the crossbar **104** so that the head rest accessory **130** extends between a fixed end **200** and a cantilevered end **202**. The fixed end **200** is positioned at the crossbar **104** where the connector **160** is inserted over the crossbar **104**. The cantilevered end **202** extends outward from the crossbar **104** past an end of the apparatus **12**. The head rest accessory **130** extends from the crossbar **104** such that the head rest accessory **130** is held in a fixed position with limited give or sway at the cantilevered end **202**. That is, the head rest accessory **130** is held substantially firmly to the crossbar **104**.

(42) As illustrated in FIG. 4, the head rest accessory **130** is coupled to the crossbar **104** so that a clearance space is provided between the frame **132** of the head rest accessory **130** and the lower frame **42** of the apparatus **12**. Accordingly, when the head rest accessory **130** is coupled to the crossbar **104**, the frame **132** of the head rest accessory **130** can extend above and across the lower frame **42** from the fixed end **200** to the cantilevered end **202**. As illustrated in FIG. 9, when the apparatus **12** is positioned in the Trendelenburg position, the frame **132** tilts and is angled downward from the fixed end **200** to the cantilevered end **202**. In this position, the space **193** in the loop configuration **190** of the frame **132** provides clearance for the lower frame **42** of the apparatus **12**. That is, the lower frame **42** of the apparatus **12** nests within the space **193** so that the head rest accessory **130** is usable with the apparatus **12** in the Trendelenburg position without the head rest accessory **130** contacting the lower frame **42**.

(43) Referring now to FIG. 5, a head rest mount **250** is coupled between the rails **134** of the accessory **130**. In the illustrated embodiment, the head rest mount **250** is coupled to the members **135** of the rails **134**. In some embodiments, the head rest mount **250** is the same as the accessory described in U.S. Pat. No. 7,520,007, which is incorporated by referenced herein in its entirety. The head rest mount **250** includes a base **252** having a pair of ends **254**. Each end **254** includes a releasable clamp **256** that can be secured to one of the rails **134**. When each clamp **256** is in an unlocked condition, the mount **250** is repositionable along the rails **134**. That is, the mount **250** is repositionable in a first dimension **258** along the longitudinal axes **140** of the rails **134**.

Additionally, the mount **250** is repositionable in the first dimension **258** along a longitudinal axis **260** of the apparatus **12**. In the first dimension **258**, the mount **250** is moveable away from or toward the crossbeam **150** of the frame **132**. That is, the mount **250** is repositionable in the first dimension between the proximal end **136** and the distal end **138** of the frame **132**. In a locked condition, the clamps **256** prevent movement of the mount **250** so that the mount **250** is locked in position relative to the rails **134**.

(44) The base **252** includes a track **262** that extends between the ends **254**. A body **264** is secured to the track **262** and repositionable along the track **262**. The body **264** includes a lock **266** that secures the body **264** to the track **262**. When the lock **266** is in an unlocked condition, the body **264** is repositionable along the track **262** in a second dimension **270** that is perpendicular to the longitudinal axes **140** of the rails **134**. The second dimension **270** is also perpendicular to the longitudinal axis **260** of the apparatus **12**. In the second dimension **270**, the body **264** is repositionable between the rails **134** so that the body **264** may be moved toward one of the rails **134** while being moved away from the other rail **134**. When the lock **266** is in a locked condition, the lock **266** prevents movement of the body **264** along the track **262**.

(45) A mount post **280** extends vertically from the body **264** and is repositionable with the body **264**. The mount post **280** extends substantially perpendicular to the longitudinal axes **140** of the rails **134**. The mount post **280** also extends substantially perpendicular to the longitudinal axis **260** of the apparatus **12**. A mount rail **282** extends substantially perpendicular from the mount post **180**. The mount rail **282** extends substantially parallel to the longitudinal axes **140** of the rails **134** and the longitudinal axis **260** of the apparatus **12**. The mount rail **282** extends in a proximal direction toward the proximal end **136** of the frame **132**.

(46) The mount rail **282** includes a body **290** that is repositionable along the mount post **280** in a third vertical dimension **292**. A lock **294** secures the body **290** to the mount post **280**. When the lock **294** is in an unlocked condition, the body **290** moves along the mount post **280** in the third dimension so that the body **290** is raised or lowered relative to the frame **132** of the accessory **130**. The third dimension **292** is perpendicular to both the first dimension **258** and the second dimension **270**. That is, the first dimension **258** and second dimensions **270** are positioned in a horizontal plane, and the third dimension **292** is positioned in a vertical plane. When the lock **294** is in a locked condition, the lock **294** prevents the body **290** from moving relative to the mount post **280**.

(47) The mount rail **282** also includes a rod **300** that extends from the body **290** substantially

perpendicular to both the longitudinal axes **140** of the rails **134** and the longitudinal axis **260** of the apparatus **12**. The rod **300** extends from the body **290** toward the proximal end **136** of the frame **132**. The rod **300** includes a cantilevered end **302**. A face rest shell **310** is coupled to the cantilevered end **302**. By moving the components of the head rest mount **250** as described above, the face rest shell **310** is movable in the first dimension **258**, the second dimension **270**, and the third dimension **292**. Additionally, the cantilevered end **302** includes a ball joint (not shown). The face rest shell **310** is coupled to the ball joint to enable the face rest shell **310** to pitch in the rotational direction **305**, yaw in the rotational direction **307**, and roll in the rotational direction **309**. The face rest shell **310** includes an upwardly facing concave surface **314**. The face rest shell **310** also includes a generally T-shaped cutout **316** to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory **130**, as illustrated in FIG. **8**. Various embodiments, of the face rest shell **310** are described in more detail below in FIGS. **10-17**.

(48) The face rest shell **310** also includes a face cushion **320** that is positioned on the face rest shell **310**, as illustrated in FIG. **6**. Referring to FIG. **7**, the face cushion **320** includes a surface **322** that is sized and shaped to the patient's face. The face cushion **320** also includes a generally T-shaped cutout **324** to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory, as illustrated in FIG. **8**.

(49) Referring to FIG. **7**, the face cushion **320** includes a top end **330** and a bottom end **332**. The top end **330** is configured to receive the patient's forehead and the bottom end **332** is configured to receive the patient's chin. An opening **334** is formed in the bottom end **332**. The cutout **324** extends from the opening **334** toward the top end **330**. The cutout **324** includes a chin section **340** having a width **342** and configured to receive the patient's chin. A cheek bone relief section **344** extends from the chin section **340** and is defined by a pair of curved contours **346** that are shaped to the patient's cheek bones. The cheek bone relief section **344** extends a width **348** that is greater than the width **342**. An eye section **350** extends from the cheek bone relief section **344** and has a width **352** that is greater than the width **348**. The eye section **350** is generally oblong in shape and provides an opening for the patient's eye sockets.

(50) Referring now to FIG. **10**, a face rest shell **400** includes an outer perimeter **402** defined by an outer edge **404**. The outer edge **404** includes a top outer edge **406** and a pair of planar side outer edges **408**. In the illustrative embodiment, the top outer edge **406** is partially circular. The top outer edge **406** is coupled to each of the side outer edges **408** by a respective curved outer edge **410**. The side outer edges **408** extend between the top outer edge **406** and a bottom outer edge **420**. The bottom outer edge **420** includes a pair of planar bottom outer edges **422**. Each of the pair of planar bottom edges **422** is coupled to a respective side outer edge **408** by an angled outer edge **424**. The planar bottom outer edges **422** extend substantially perpendicular to the pair of side outer edges **408**.

(51) An opening **430** is formed in the bottom outer edge **420**. That is, the opening **430** is formed between the pair of planar bottom outer edges **422**. A cutout **432** extends from the opening **430** toward the top outer edge **406**. The cutout **432** is generally T-shaped and defined by an inner edge **434**. The cutout **432** includes a mouth cutout **436** that extends from the opening **430**. An eye cutout **438** extends from the mouth cutout **436** toward the top outer edge **406**. A downwardly facing convex surface **440** extends between the outer edge **404** and the inner edge **434**.

(52) Referring to FIG. **11**, the face rest shell **400** includes an upwardly facing concave surface **450** extending between the outer edge **404** and the inner edge **434**. The mouth cutout **436** includes a pair of side mouth inner edges **452** extending from the opening **430** to a cheek rest **454** that flares inward. The pair of side mouth inner edges **452** each include a plurality of side mouth inner edges **464**, wherein at least one of the plurality of side mouth inner edges **464** is substantially arcuate. A cheek rest opening **456** is defined between an inwardly most point **458** of each of the cheek rests **454**. In the illustrated embodiment, the cheek rest opening **456** has a width **460** that is less than a

width **462** of the opening **430**.

(53) The eye cutout **438** extends from the mouth cutout **436**. The eye cutout **438** extends from the cheek rest opening **456**. The eye cutout **438** includes a pair of curved inner edges **470** that extend from the cheek rest opening **456** to a planar top inner edge **472**. The pair of curved inner edge **470** are substantially arcuate. The top inner edge **472** extends substantially perpendicular to the side outer edges **408**. The top inner edge **472** also extends substantially parallel to the bottom outer edge **420**. Each cheek rest **454** is defined between the mouth cutout **436** and the eye cutout **438** so that each cheek rest **454** is defined at least partially by a respective side mouth inner edge **452** of the mouth cutout **436** and at least partially by a respective curved inner edge **470** of the eye cutout **438**.

(54) A top wall **401** extends from the top outer edge **406**. In the illustrative embodiment, the top wall **401** extends upwardly from the upwardly facing concave surface **450**. A pair of side walls **403** extend from the each of the planar side outer edges **408**. In the illustrative embodiment, each of the pair of side walls **403** extends upwardly from the upwardly facing concave surface **450**. The top wall **401** and the side walls **403** collectively retain the face cushion **320** on the upwardly facing concave surface **450**.

(55) Referring now to FIG. **12**, a face rest shell **500** includes an outer perimeter **502** defined by an outer edge **504**. The outer edge **504** includes a top outer edge **506** and a pair of planar side outer edges **508**. In the illustrative embodiment, the top outer edge **506** is partially circular. The top outer edge **506** is coupled to each of the side outer edges **508** by a respective curved outer edge **510**. The side outer edges **508** extend between the top outer edge **506** and a bottom outer edge **520**. The bottom outer edge **520** includes a pair of planar bottom outer edges **522**. Each of the pair of planar bottom edges **522** is coupled to a respective side outer edge **508** by an angled outer edge **524**. The planar bottom outer edges **522** extend substantially perpendicular to the pair of side outer edges **508**.

(56) An opening **530** is formed in the bottom outer edge **520**. That is, the opening **530** is formed between the pair of planar bottom outer edges **522**. A cutout **532** extends from the opening **530** toward the top outer edge **506**. The cutout **532** is generally T-shaped and defined by an inner edge **534**. The cutout **532** includes a mouth cutout **536** that extends from the opening **530**. An eye cutout **538** extends from the mouth cutout **536** toward the top outer edge **506**. A downwardly facing convex surface **540** extends between the outer edge **504** and the inner edge **534**.

(57) Referring to FIG. **13**, the face rest shell **500** includes an upwardly facing concave surface **550** extending between the outer edge **504** and the inner edge **534**. The mouth cutout **536** includes a pair of side mouth inner edges **552** extending from the opening **530** to a cheek rest **554** that flares inward. The pair of side mouth inner edges **552** each include a plurality of side mouth inner edges **564**, wherein at least one of the plurality of side mouth inner edges **564** is substantially circular. A cheek rest opening **556** is defined between an inwardly most point **558** of each of the cheek rests **554**. In the illustrated embodiment, the cheek rest opening **556** has a width **560** that is substantially equal to a width **562** of the opening **530**.

(58) The eye cutout **538** extends from the mouth cutout **536**. The eye cutout **538** extends from the cheek rest opening **556**. The eye cutout **538** includes a pair of curved inner edges **570** that extend from the cheek rest opening **556** to a planar top inner edge **572**. The pair of curved inner edge **570** are substantially circular. The top inner edge **572** extends substantially perpendicular to the side outer edges **508**. The top inner edge **572** also extends substantially parallel to the bottom outer edge **520**. Each cheek rest **554** is defined between the mouth cutout **536** and the eye cutout **538** so that each cheek rest **554** is defined at least partially by a respective side mouth inner edge **552** of the mouth cutout **536** and at least partially by a respective curved inner edge **570** of the eye cutout **538**.

(59) A top wall **501** extends from the top outer edge **506**. In the illustrative embodiment, the top wall **501** extends upwardly from the upwardly facing concave surface **550**. A pair of side walls **503** extend from the each of the planar side outer edges **508**. In the illustrative embodiment, each of the pair of side walls **503** extends upwardly from the upwardly facing concave surface **550**. The top

wall **501** and the side walls **503** collectively retain the face cushion **520** on the upwardly facing concave surface **550**.

(60) Referring now to FIG. **14**, a face rest shell **600** includes an outer perimeter **602** defined by an outer edge **604**. The outer edge **604** includes a top outer edge **606** and a pair of planar side outer edges **608**. In the illustrative embodiment, the top outer edge **606** is partially circular. The top outer edge **606** is coupled to each of the side outer edges **608** by a respective curved outer edge **610**. The side outer edges **608** extend between the top outer edge **606** and a bottom outer edge **620**. The bottom outer edge **620** includes a pair of planar bottom outer edges **622**. Each of the pair of planar bottom edges **622** is coupled to a respective side outer edge **608** by an angled outer edge **624**. The planar bottom outer edges **622** extend substantially perpendicular to the pair of side outer edges **608**.

(61) An opening **630** is formed in the bottom outer edge **620**. That is, the opening **630** is formed between the pair of planar bottom outer edges **622**. A cutout **632** extends from the opening **630** toward the top outer edge **606**. The cutout **632** is generally T-shaped and defined by an inner edge **634**. The cutout **632** includes a mouth cutout **636** that extends from the opening **630**. An eye cutout **638** extends from the mouth cutout **636** toward the top outer edge **606**. A downwardly facing convex surface **640** extends between the outer edge **604** and the inner edge **634**.

(62) Referring to FIG. **15**, the face rest shell **600** includes an upwardly facing concave surface **650** extending between the outer edge **604** and the inner edge **634**. The mouth cutout **636** includes a pair of side mouth inner edges **652** extending from the opening **630** to a cheek rest **654** that flares inward. The pair of side mouth inner edges **652** each include a plurality of side mouth inner edges **664**, wherein at least one of the plurality of side mouth inner edges **664** is substantially circular. A cheek rest opening **656** is defined between an inwardly most point **658** of each of the cheek rests **654**. In the illustrated embodiment, the cheek rest opening **656** has a width **660** that is substantially equal to a width **662** of the opening **630**.

(63) The eye cutout **638** extends from the mouth cutout **636**. The eye cutout **638** extends from the cheek rest opening **656**. The eye cutout **638** includes a pair of curved inner edges **670** that extend from the cheek rest opening **656** to a planar top inner edge **672**. The pair of curved inner edge **670** are substantially circular. The top inner edge **672** extends substantially perpendicular to the side outer edges **608**. The top inner edge **672** also extends substantially parallel to the bottom outer edge **620**. Each cheek rest **654** is defined between the mouth cutout **636** and the eye cutout **638** so that each cheek rest **654** is defined at least partially by a respective side mouth inner edge **652** of the mouth cutout **636** and at least partially by a respective curved inner edge **670** of the eye cutout **638**.

(64) Each cheek rest **654** includes a cheek rest relief surface **680** formed in the upwardly facing concave surface **650**. The cheek rest relief surface **680** is a concave surface having a radius of curvature that differs from a radius of curvature of the upwardly facing concave surface **650**. A perimeter **682** of each cheek rest relief surface **680** is defined by a portion of the respective curved inner edge **670** of the eye cutout **638** and a curved bottom relief edge **684**. The cheek rest relief surface **680** is configured to reduce pressure on the patient's cheeks when the patient is positioned in a prone position on the face rest shell **600**.

(65) A top wall **601** extends from the top outer edge **606**. In the illustrative embodiment, the top wall **601** extends upwardly from the upwardly facing concave surface **650**. A pair of side walls **603** extend from the each of the planar side outer edges **608**. In the illustrative embodiment, each of the pair of side walls **603** extends upwardly from the upwardly facing concave surface **650**. The top wall **601** and the side walls **603** collectively retain the face cushion **620** on the upwardly facing concave surface **650**.

(66) Referring now to FIG. **16**, a face rest shell **700** includes an outer perimeter **702** defined by an outer edge **704**. The outer edge **704** includes a top outer edge **706** and a pair of planar side outer edges **708**. In the illustrative embodiment, the top outer edge **706** is partially circular. The top outer edge **706** is coupled to each of the side outer edges **708** by a respective curved outer edge **710**. The

side outer edges **708** extend between the top outer edge **706** and a bottom outer edge **720**. The bottom outer edge **720** includes a pair of planar bottom outer edges **722**. Each of the pair of planar bottom edges **722** is coupled to a respective side outer edge **708** by an angled outer edge **724**. The planar bottom outer edges **722** extend substantially perpendicular to the pair of side outer edges **708**.

(67) An opening **730** is formed in the bottom outer edge **720**. That is, the opening **730** is formed between the pair of planar bottom outer edges **722**. A cutout **732** extends from the opening **730** toward the top outer edge **706**. The cutout **732** is generally T-shaped and defined by an inner edge **734**. The cutout **732** includes a mouth cutout **736** that extends from the opening **730**. An eye cutout **738** extends from the mouth cutout **736** toward the top outer edge **706**. A downwardly facing convex surface **740** extends between the outer edge **704** and the inner edge **734**.

(68) Referring to FIG. **17**, the face rest shell **700** includes an upwardly facing concave surface **750** extending between the outer edge **704** and the inner edge **734**. The mouth cutout **736** includes a pair of side mouth inner edges **752** extending from the opening **730** to a cheek rest **754** that flares inward. The pair of side mouth inner edges **752** each include a plurality of side mouth inner edges **764**, wherein at least one of the plurality of side mouth inner edges **764** is substantially planar. A cheek rest opening **756** is defined between an inwardly most point **758** of each of the cheek rests **754**. In the illustrated embodiment, the cheek rest opening **756** has a width **760** that is less than a width **762** of the opening **730**.

(69) The eye cutout **738** extends from the mouth cutout **736**. The eye cutout **738** extends from the cheek rest opening **756**. The eye cutout **738** includes a pair of curved inner edges **770** that extend from the cheek rest opening **756** to a planar top inner edge **772**. The pair of curved inner edge **770** are at least partially circular. The top inner edge **772** extends substantially perpendicular to the side outer edges **708**. The top inner edge **772** also extends substantially parallel to the bottom outer edge **720**. Each cheek rest **754** is defined between the mouth cutout **736** and the eye cutout **738** so that each cheek rest **754** is defined at least partially by a respective side mouth inner edge **752** of the mouth cutout **736** and at least partially by a respective curved inner edge **770** of the eye cutout **738**.

(70) A top wall **701** extends from the top outer edge **706**. In the illustrative embodiment, the top wall **701** extends upwardly from the upwardly facing concave surface **750**. A pair of side walls **703** extends from the each of the planar side outer edges **708**. In the illustrative embodiment, each of the pair of side walls **703** extends upwardly from the upwardly facing concave surface **750**. The top wall **701** and the side walls **703** collectively retain the face cushion **720** on the upwardly facing concave surface **750**.

(71) Referring to FIG. **18**, a patient support apparatus **1000** is illustrated having a head rest accessory **1002** extending therefrom. A face rest shell **1006** is coupled to an end **1008** of the frame **1002**. In some embodiments, the patient support apparatus **1000** may be the patient support apparatus **12**. The head rest accessory **1002** may be the head rest accessory **130**, in some embodiments. Additionally, in some embodiments, the face rest shell **1006** may be any one of face rest shells **310**, **400**, **500**, **600**, or **700**. FIG. **18** illustrates a head end **1020** of the patient support apparatus **1000**, wherein the face rest shell **1006** extends from the head end **1020**. A patient **1022** is positioned in a prone position on the patient support apparatus **1000**. The patient's head **1024** is positioned in the face rest shell **1006**.

(72) A lateral axis **1030** extends between a first side **1032** and a second side **1034** of the patient support apparatus **1000**. The face rest shell **1006** is configured to be positioned relative to the lateral axis **1030**. That is, as described above, the face rest shell **1006** may be rolled relative to the lateral axis **1030**. In the illustrated embodiment, the face rest shell **1006** is shown rolled to a lateral angle **1040**. The lateral angle **1040** may be any suitable angle greater than 0 degrees that accommodates a morphology of the patient **1022** when the patient's head **1024** is positioned in the face rest shell **1006**. Altering the lateral angle **1040** facilitates improving patient comfort and reducing pressure sores on the patient. A caregiver may alter the lateral angle **1040** before the

patient's head **1024** is positioned in the face rest shell **1006** utilizing sensors, as described in more detail below. Additionally, the lateral angle **1040** may be further adjusted after the patient's head **1024** is positioned in the face rest shell **1006** utilizing the sensors described below. Although FIG. **18** illustrates the face rest shell **1006** rolled in one direction, it will be appreciated that the face rest shell **1006** may be rolled in either direction.

(73) Referring now to FIG. **19**, the patient support apparatus **1000** extends between the head end **1020** and a foot end **1050**. The patient **1022** is positioned on the patient support apparatus **1000** in a prone position between the head end **1020** and the foot end **1050**. The patient's head **1024** extends from the head end **1020** of the patient support apparatus **1000** and is positioned in the face rest shell **1006**.

(74) A longitudinal axis **1052** extends between the head end **1020** and the foot end **1050** of the patient support apparatus **1000**. The face rest shell **1006** is configured to be positioned relative to the longitudinal axis **1052**. That is, as described above, the face rest shell **1006** may be pitched relative to the longitudinal axis **1052**. In the illustrated embodiment, the face rest shell **1006** is shown pitched to a longitudinal angle **1054**. The longitudinal angle **1054** may be any suitable angle greater than 0 degrees that accommodates a morphology of the patient **1022** when the patient's head **1024** is positioned in the face rest shell **1006**. Altering the longitudinal angle **1054** facilitates improving patient comfort and reducing pressure sores on the patient. A caregiver may alter the longitudinal angle **1054** before the patient's head **1024** is positioned in the face rest shell **1006** utilizing sensors, as described in more detail below. Additionally, the longitudinal angle **1054** may be further adjusted after the patient's head **1024** is positioned in the face rest shell **1006** utilizing the sensors described below. Although FIG. **19** illustrates the face rest shell **1006** pitch in one direction, it will be appreciated that the face rest shell **1006** may be pitched in either direction.

(75) Referring to FIG. **20**, the face rest shell **1006** includes a head end **1100** and an opposite chin end **1102**. A pair of sides **1104** extends between the head end **1100** and the chin end **1102**. A centerline **1106** that is substantially equidistant from each side **1104** extends substantially parallel to the sides **1104** between the head end **1100** and the chin end **1102**. An opening **1108** is formed in the chin end **1102**. A generally T-shaped cutout **1110** extends from the opening **1108** toward the head end **1100**. The generally T-shaped cutout **1110** includes chin cutout **1112** that extends from the opening **1108**. A nose cutout **1114** extends from the chin cutout **1112** toward the head end **1100**, and an eye cutout **1116** extends from the nose cutout **1114** toward the head end **1100**.

(76) A lateral sensor **1120** is positioned on the head end **1100** and aligned with the centerline **1106**. The lateral sensor **1120** extends across the head end **1100** between the sides **1104**. The lateral sensor **1120** is configured to measure the lateral angle **1040** of the face rest shell **1006**. A longitudinal sensor **1122** is positioned on one of the sides **1104**. It will be appreciated that the longitudinal sensor **1122** may be positioned on either of the sides **1104** or both of the sides **1104**. The longitudinal sensor **1122** is substantially aligned with the eye cutout **1116**, in the illustrated embodiment. In some embodiments, the longitudinal sensor **1122** may be aligned with the nose cutout **1114** or the chin cutout **1112**. The longitudinal sensor **1122** is configured to measure the longitudinal angle **1054** of the face rest shell **1006**.

(77) Referring now to FIG. **21**, the lateral sensor **1120** is configured as a spirit level vial **1140** having fluid **1142** retained therein. A bubble **1144** is present in the fluid **1142** and configured to move through the fluid **1142** dependent on the lateral angle **1040** of the face rest shell **1006** and the lateral sensor **1120**. A plurality of indicia **1146** are marked on the vial **1140** to indicate a measurement of the lateral angle **1040**. For example, if the bubble **1144** is positioned between a pair of center indicia **1148**, the lateral sensor **1120** indicates that the lateral angle **1040** is substantially zero. A degree of the roll of the face rest shell **1006** is indicated by an amount to which the bubble **1144** floats outside of the center indicia **1148**. It will be appreciated, that additional indicia including angle measurements, e.g. 5 degrees, 10 degrees, etc., may be provided on the head end **1100** or the vial **1140** to provide a measurement of the lateral angle **1140**.

(78) Referring to FIG. 22, the longitudinal sensor **1122** is configured as a spirit level vial **1150** having fluid **1152** retained therein. A bubble **1154** is present in the fluid **1152** and configured to move through the fluid **1152** dependent on the longitudinal angle **1054** of the face rest shell **1006** and the longitudinal sensor **1122**. A plurality of indicia **1156** are marked on the vial **1150** to indicate a measurement of the longitudinal angle **1054**. For example, if the bubble **1154** is positioned between a pair of center indicia **1158**, the longitudinal sensor **1122** indicates that the longitudinal angle **1054** is substantially zero. A degree of the pitch of the face rest shell **1006** is indicated by an amount to which the bubble **1154** floats outside of the center indicia **1158**. It will be appreciated, that additional indicia including angle measurements, e.g. 5 degrees, 10 degrees, etc., may be provided on the head end **1100** or the vial **1150** to provide a measurement of the longitudinal angle **1054**.

(79) FIG. 23 illustrates another embodiment of the face rest shell **1006** having a lateral sensor **1200** positioned on the head end **1100** and aligned with the centerline **1106**. The lateral sensor **1200** extends across the head end **1100** between the sides **1104**. The lateral sensor **1200** is configured to measure the lateral angle **1040** of the face rest shell **1006**. A longitudinal sensor **1202** is positioned on one of the sides **1104**. It will be appreciated that the longitudinal sensor **1202** may be positioned on either of the sides **1104** or both of the sides **1104**. The longitudinal sensor **1202** is substantially aligned with the eye cutout **1116**, in the illustrated embodiment. In some embodiments, the longitudinal sensor **1202** may be aligned with the nose cutout **1114** or the chin cutout **1112**. The longitudinal sensor **1202** is configured to measure the longitudinal angle **1054** of the face rest shell **1006**.

(80) Referring now to FIG. 24, the lateral sensor **1200** is configured as a spirit level vial **1210** having fluid **1212** retained therein. A bubble **1214** is present in the fluid **1212** and configured to move through the fluid **1212** dependent on the lateral angle **1040** of the face rest shell **1006** and the lateral sensor **1200**. A plurality of indicia **1216** are marked on the vial **1210** to indicate a measurement of the lateral angle **1040**. For example, if the bubble **1214** is positioned between a pair of center indicia **1218**, the lateral sensor **1200** indicates that the lateral angle **1040** is substantially zero. A degree of the roll of the face rest shell **1006** is indicated by an amount to which the bubble **1214** floats outside of the center indicia **1218**. It will be appreciated, that additional indicia including angle measurements, e.g. 5 degrees, 10 degrees, etc., may be provided on the head end **1100** or the vial **1210** to provide a measurement of the lateral angle **1140**.

(81) Referring to FIG. 25, the longitudinal sensor **1202** is configured as a liquid level **1230** that includes an artificial horizon inclinometer **1232**. The longitudinal sensor **1202** includes a transparent case **1234** filled with a fluid **1236**. A top surface **1238** of the fluid **1236** is configured to measure the longitudinal angle **1054**. That is, the top surface **1238** of the fluid **1236** remains substantially horizontal while the transparent case **1234** pitches with the face rest shell **1006**. When the face rest shell **1006** is pitched, the top surface **1238** of the fluid **1236** aligns with indicia **1240** on the transparent case **1234** or on the face rest shell **1006** around the transparent case **1234** to indicate the longitudinal angle **1054**.

(82) FIG. 26 illustrates another embodiment of the face rest shell **1006** having a lateral sensor **1300** positioned on the head end **1100** and aligned with the centerline **1106**. The lateral sensor **1300** extends across the head end **1100** between the sides **1104**. The lateral sensor **1300** is configured to measure the lateral angle **1040** of the face rest shell **1006**. A longitudinal sensor **1302** is positioned on one of the sides **1104**. It will be appreciated that the longitudinal sensor **1302** may be positioned on either of the sides **1104** or both of the sides **1104**. The longitudinal sensor **1302** is substantially aligned with the eye cutout **1116**, in the illustrated embodiment. In some embodiments, the longitudinal sensor **1302** may be aligned with the nose cutout **1114** or the chin cutout **1112**. The longitudinal sensor **1302** is configured to measure the longitudinal angle **1054** of the face rest shell **1006**.

(83) Referring to FIG. 27, the lateral sensor **1300** is configured as a liquid level **1310** that includes

an artificial horizon inclinometer **1312**. The lateral sensor **1300** includes a transparent case **1314** filled with a fluid **1316**. A top surface **1318** of the fluid **1316** is configured to measure the lateral angle **1040**. That is, the top surface **1318** of the fluid **1316** remains substantially horizontal while the transparent case **1314** rolls with the face rest shell **1006**. When the face rest shell **1006** is rolled, the top surface **1318** of the fluid **1316** aligns with indicia **1320** on the transparent case **1314** to indicate the lateral angle **1040**. It will be appreciated that, since the transparent case **1314** is spherical and three-dimensional, the lateral sensor **1300** may also double as a longitudinal sensor. Specifically, the top surface **1318** of the fluid **1316** also remains substantially horizontal while the transparent case **1314** pitches with the face rest shell **1006**. When the face rest shell **1006** is pitched, the top surface **1318** of the fluid **1316** aligns with indicia on the transparent case **1314** to indicate the longitudinal angle **1054**.

(84) Referring now to FIG. **28**, the longitudinal sensor **1302** is configured as a liquid level **1350** that includes an artificial horizon inclinometer **1352**. The longitudinal sensor **1302** includes a transparent case **1354** filled with a fluid **1356**. A top surface **1358** of the fluid **1356** is configured to measure the longitudinal angle **1054**. That is, the top surface **1358** of the fluid **1356** remains substantially horizontal while the transparent case **1354** pitches with the face rest shell **1006**. When the face rest shell **1006** is pitched, the top surface **1358** of the fluid **1356** aligns with indicia **1360** on the transparent case **1354** or on the face rest shell **1006** around the transparent case **1354** to indicate the longitudinal angle **1054**.

(85) It will be appreciated that the face rest shell **1006** may use any combination of the lateral sensors **1120**, **1200**, **1300** and longitudinal sensors **1122**, **1202**, **1302**. The lateral sensors **1120**, **1200**, **1300** and longitudinal sensors **1122**, **1202**, **1302** described above provide a caregiver a visible and intuitive tilt indicators system on the face rest shell **1006** to easily set a correct position of the face rest shell **1006** according the patient morphology. The lateral sensors **1120**, **1200**, **1300** and longitudinal sensors **1122**, **1202**, **1302** facilitate visual management of the tilt angles of the face rest shell **1006** to allow the adjustment of the face rest shell **1006** to prevent and reduce skin injury risks and interface pressure on the patient's face. Efficiency is provided to the caregiver by pre-setting the face rest shell **1006** and allowing ease of use of the face rest shell **1006** during the prone procedure. The lateral sensors **1120**, **1200**, **1300** and longitudinal sensors **1122**, **1202**, **1302** provide the caregiver an indication of the tilt angle of the face rest shell **1006** relative to both the longitudinal axis **1052** and the lateral axis **1030**. In some embodiments, the face rest shell **1006** may be adjusted before installation of the face rest shell **1006** on the head rest accessory **1002**.

(86) Before fixing the face rest shell **1006** on the patient support apparatus **1000**, the caregiver can pre-set the lateral angle **1040** and longitudinal angle **1054** of the face rest shell **1006** to avoid adjusting the lateral angle **1040** and longitudinal angle **1054** with the patient's head **1024** in the face rest shell **1006**. This pre-setting protocol facilitates saving time and minimizes risks during the patient prone positioning process. Depending on patient morphology and caregiver observation, the caregiver can decide to adjust the lateral angle **1040** and longitudinal angle **1054** of the face rest shell to provide the best ergonomic position for the patient **1022**.

(87) Referring now to FIG. **29**, a face rest shell **1400** includes an outer perimeter **1402** defined by an outer edge **1404**. The outer edge **1404** includes a top outer edge **1406** and a pair of planar side outer edges **1408**. In the illustrative embodiment, the top outer edge **1406** is partially circular. The top outer edge **1406** is coupled to each of the side outer edges **1408** by a respective curved outer edge **1410**. The side outer edges **1408** extend between the top outer edge **1406** and a bottom outer edge **1420**. The bottom outer edge **1420** includes a pair of planar bottom outer edges **1422**. Each of the pair of planar bottom edges **1422** is coupled to a respective side outer edge **1408** by an angled outer edge **1424**. The planar bottom outer edges **1422** extend substantially perpendicular to the pair of side outer edges **1408**.

(88) An opening **1430** is formed in the bottom outer edge **1420**. That is, the opening **1430** is formed between the pair of planar bottom outer edges **1422**. A cutout **1432** extends from the opening **1430**

toward the top outer edge **1406**. The cutout **1432** is generally T-shaped and defined by an inner edge **1434**. The cutout **1432** includes a mouth cutout **1436** that extends from the opening **1430**. An eye cutout **1438** extends from the mouth cutout **1436** toward the top outer edge **1406**. A downwardly facing convex surface **1440** extends between the outer edge **1404** and the inner edge **1434**.

(89) A top wall **1490** extends from the top outer edge **1406**. In the illustrative embodiment, the top wall **1490** extends upwardly from the downwardly facing convex surface **1440**. A pair of side walls **1492** extends from the each of the planar side outer edges **1408**. In the illustrative embodiment, each of the pair of side walls **1492** extends upwardly from the downwardly facing convex surface **1440**.

(90) Referring to FIG. 30, the face rest shell **1400** includes an upwardly facing concave surface **1450** extending between the outer edge **1404** and the inner edge **1434**. The mouth cutout **1436** includes a pair of side mouth inner edges **1452** extending from the opening **1430** to a cheek rest **1454** that flares inward. The pair of side mouth inner edges **1452** each include a pair of planar side mouth inner edges **1464**. A cheek rest opening **1456** is defined between an inwardly most point **1458** of each of the cheek rests **1454**. In the illustrated embodiment, the cheek rest opening **1456** has a width **1460** that is less than a width **1462** of the opening **1430**.

(91) The eye cutout **1438** extends from the mouth cutout **1436**. The eye cutout **1438** extends from the cheek rest opening **1456**. The eye cutout **1438** includes a pair of curved inner edges **1470** that extend from the cheek rest opening **1456** to a planar top inner edge **1472**. The pair of curved inner edges **1470** are substantially circular. Each cheek rest **1454** is defined between the mouth cutout **1436** and the eye cutout **1438** so that each cheek rest **1454** is defined at least partially by a respective side mouth inner edge **1452** of the mouth cutout **1436** and at least partially by a respective curved inner edge **1470** of the eye cutout **1438**. The eye cutout **1438** has a width **1498** that is greater than the widths **1460** and **1462**.

(92) A relief surface **1480** is formed in the upwardly facing concave surface **1450**. The relief surface **1480** extends along the cheek rest **1454** and around the eye cutout **1438**. The relief surface **1480** is configured to reduce pressure on the patient's cheeks and around the patient's eyes when the patient is positioned in a prone position on the face rest shell **600**. The relief surface **1480** is configured to distribute interface pressure from protruding facial features of the patient.

(93) The top wall **1490** extends upwardly from the upwardly facing concave surface **1450**. Each of the pair of side walls **1492** extends upwardly from the upwardly facing concave surface **1450**. The top wall **1490** and the side walls **1492** collectively retain the face cushion **1420** on the upwardly facing concave surface **1450**.

(94) Referring now to FIG. 31, a face cushion **1500** includes an outer perimeter **1502**. The outer perimeter **1502** includes a top outer edge **1504** that is substantially circular. A pair of side outer edges **1506** extend from the top outer edge **1504** to a pair of angled bottom outer edges **1508**. That is, the angled bottom outer edges **1508** extend non-orthogonal to the side outer edges **1506**. In some embodiments, the bottom outer edges **1508** are not angled and extend substantially perpendicular to the side outer edges **1506**. An angled outer edge **1510** extends between each bottom outer edge **1508** and the respective side outer edge **1506**.

(95) The face cushion **1500** includes a top end **1530**, adjacent the top outer edge **1504**, and a bottom end **1532**, adjacent the bottom outer edge **1508**. The top end **1530** is configured to receive the patient's forehead and the bottom end **1532** is configured to receive the patient's jaw. An opening **1534** is formed in the bottom end **1532** and extends inward from the bottom outer edge **1508** toward the top outer edge **1504**. A pair of chin sections **1540** are configured to receive the patient's chin. A pair of cheek bone relief sections **1542** are shaped to support the patient's cheek bones.

(96) The chin sections **1540** are defined by a pair of substantially circular inner edges **1560** extending from the opening **1534**. An opening **1580** between the chin sections **1540** has a width

1582. A mouth cutout **1562** extends from the opening **1580** and is defined by a pair of planar mouth side inner edges **1564** extending from the inner edges **1560**. The mouth side inner edges **1564** flare inward from the inner edges **1560**. A nose cutout **1566** is defined between an inwardly most point **1568** of each of the cheek bone relief sections **1542**. In the illustrated embodiment, the nose cutout **1566** has a width **1570** that is substantially equal to the width **1582** of the opening **1580**.

(97) An eye cutout **1584** extends from the mouth cutout **1562**. The eye cutout **1584** extends from the nose cutout **1566**. The eye cutout **1584** includes a pair of planar bottom edges **1586** that extend from the nose cutout **1566**. A pair of curved inner edges **1590** extend from the pair of planar bottom edges **1586** to a curved top inner edge **1572**. The pair of curved inner edge **1590** are substantially circular. The cheek bone relief sections **1542** are defined between the mouth cutout **1562** and the eye cutout **1584** so the cheek bone relief sections **1542** are defined at least partially the mouth cutout **1562** and at least partially by the eye cutout **1584**. The eye cutout **1584** has a width **1596** that is greater than the widths **1582** and **1570**.

(98) Any theory, mechanism of operation, proof, or finding stated herein is meant to further enhance understanding of principles of the present disclosure and is not intended to make the present disclosure in any way dependent upon such theory, mechanism of operation, illustrative embodiment, proof, or finding. It should be understood that while the use of the word preferable, preferably or preferred in the description above indicates that the feature so described can be more desirable, it nonetheless cannot be necessary and embodiments lacking the same can be contemplated as within the scope of the disclosure, that scope being defined by the claims that follow.

(99) In reading the claims it is intended that when words such as “a,” “an,” “at least one,” “at least a portion” are used there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. When the language “at least a portion” and/or “a portion” is used, the item can include a portion and/or the entire item unless specifically stated to the contrary.

(100) It should be understood that only selected embodiments have been shown and described and that all possible alternatives, modifications, aspects, combinations, principles, variations, and equivalents that come within the spirit of the disclosure as defined herein or by any of the following claims are desired to be protected. While embodiments of the disclosure have been illustrated and described in detail in the drawings and foregoing description, the same are to be considered as illustrative and not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Additional alternatives, modifications and variations can be apparent to those skilled in the art. Also, while multiple inventive aspects and principles have been presented, they need not be utilized in combination, and many combinations of aspects and principles are possible in light of the various embodiments provided above.

Claims

1. A head rest accessory for coupling to a frame member at a head end of a pivotable torso support section of a hospital bed, the head rest accessory comprising a frame having a pair of rails that are spaced apart and substantially parallel and that extend from a distal end to a proximal end, a crossbeam coupled to the distal ends of the pair of rails, a U-shaped connector that is coupled to the proximal ends of a respective rail of the pair of rails, the U-shaped connector opening downwardly to permit the U-shaped connector to be insertable downwardly over the frame member to couple the frame to the pivotable torso support section so that the frame is cantilevered from the frame member, a head rest mount coupled to the frame and extending between the pair of rails, the head rest mount having at least one releasable clamp that grips a corresponding one of the pair of rails to lock the head rest mount in place on the frame when the releasable clamp is in a locked condition, and the head rest mount being repositionable along the pair of rails when the releasable clamp is in an unlocked condition, and a head rest coupled to the head rest mount, the head rest including a

face rest shell having an upwardly facing concave surface and having a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory, wherein the generally T-shaped cutout includes a chin cutout including an open space configured to receive the patient's central chin region, wherein the face shell includes a pair of cheek rests that flare inwardly to define a cheek rest opening between an inward most point of each of the cheek rests, wherein a cheek rest relief surface is formed in each cheek rest in communication with an eye cutout to facilitate distributing interface pressure from protruding facial features of the patient, wherein each cheek rest relief surface includes a concave surface having a radius of curvature that differs from a radius of curvature of the respective cheek rest, wherein each of the pair of rails includes a loop configured to position over a base frame of the bed when the bed is in a Trendelenburg position, wherein: the U-shaped connector includes a pair of sockets; and each loop includes a post configured to be inserted into a respective socket, wherein the posts are removably inserted into each of the pair of sockets.

2. The head rest accessory of claim 1, wherein the T-shaped cutout extends from an opening that defines the chin cutout to a mouth cutout having a width that is greater than a width of the chin cutout, wherein an inner edge of the T-shaped cutout flares inwardly between the chin cutout and the mouth cutout, wherein the T-shaped cutout extends from the mouth cutout to the eye cutout which has a width that is greater than the width of the mouth cutout, wherein the inner edge of the T-shaped cutout flares inwardly between the mouth cutout and the eye cutout.

3. The head rest accessory of claim 1, wherein the crossbeam extends orthogonal to each of the pair of rails.

4. The head rest accessory of claim 1, wherein the U-shaped connector extends perpendicular to the frame and includes a pair of flanges and a slot extending between the pair of flanges, wherein each of the pair of flanges includes a planar surface, wherein the slot is defined by the planar surfaces of the pair of flanges.

5. The head rest accessory of claim 1, wherein each of the pair of rails has a rectangular cross-section taken in a vertical plane.

6. The head rest accessory of claim 1, wherein at least a portion of the U-shaped connector includes at least one of a plastic and composite.

7. The head rest accessory of claim 1, wherein the head rest mount is repositionable along the pair of rails along a longitudinal axis of the hospital bed.

8. The head rest accessory of claim 1, wherein the head rest mount includes a base that is repositionable along the pair of rails.

9. The head rest accessory of claim 8, wherein the base includes a track and a body repositionable along the track in a direction perpendicular to the longitudinal axis of the hospital bed.

10. The head rest accessory of claim 9, further comprising: a post extending vertically from the body, and a mount rail extending from the post and repositionable vertically along the post, wherein the head rest is coupled to a cantilevered end of the rail.

11. The head rest accessory of claim 10, wherein: the head rest is rotatable about the mount rail in a first rotational direction to roll the head rest, the head rest is rotatable about the mount rail in a second rotational direction to pitch the head rest, and the head rest is rotatable about the mount rail in a third rotational direction to yaw the head rest.

12. The head rest accessory of claim 1, wherein the face rest shell includes: a downwardly facing convex surface opposite the upwardly facing concave surface, an outer perimeter having a partially circular top outer edge and a bottom outer edge, the outer perimeter including a pair of planar side outer edges extending between the top outer edge and the bottom outer edge, and a bottom opening formed in the bottom outer edge, wherein the generally T-shaped cutout extends from the bottom opening, the generally T-shaped cutout defined by an inner edge, the generally T-shaped cutout including a mouth cutout extending from the opening and an eye cutout extending from the mouth

cutout, wherein the eye cutout is defined by a planar top inner edge and a pair of curved inner edges extending from the planar top inner edge, wherein the concave surface and the convex surface extend between the outer edge of the cutout and the outer perimeter.

13. The head rest accessory of claim 12, wherein the face rest shell includes: a top wall extending from the top outer edge, and a pair of side walls, wherein each of the pair of side walls extends from one of the pair of planar side outer edges.

14. The head rest accessory of claim 13, further comprising a face cushion is positioned on the concave surface of the face rest shell and retained in the concave surface of the face rest shell by the top wall and the pair of side walls.

15. The head rest accessory of claim 14, wherein the face cushion includes a surface that is sized and shaped to the patient's face, wherein the face cushion includes a generally T-shaped cutout to receive a patient's eyes, nose, mouth, and a central chin region of the patient's face when the patient is in a prone position on the head rest accessory.

16. The head rest accessory of claim 1, wherein the face rest shell includes at least one angle sensor attached to the face rest shell to indicate a tilt angle of the face rest shell.

17. The head rest accessory of claim 16, wherein the at least one angle sensor is positioned along a side of the face rest shell and aligned with an eye cutout of the generally T-shaped cutout, wherein the at least one angle sensor measures a longitudinal tilt angle of the face rest shell.

18. The head rest accessory of claim 16, wherein the at least one angle sensor is positioned on a head end of the face rest shell on a centerline of the face rest shell extending between the head end and a chin end of the face rest shell, wherein the at least one angle sensor measures a lateral tilt angle of the face rest shell.
